

CASE STUDY

MULTI-HOSPITAL DICOM SHARING PLATFORM

HIPAA-aware healthcare platform for real-time DICOM image sync across hospitals.

Category: fullstack

Tech: React.js, Firebase, MySQL, DICOM

PROBLEM & SOLUTION

REPLACING MANUAL TRANSFERS

Hospitals shared DICOM images via USB/email. This platform digitizes the workflow with real-time sync across nodes.

- * Nodes: 5+

- * Latency Cut: 35%

- * Security: HIPAA-Aware

CASE STUDY

MULTI-HOSPITAL DICOM SHARING PLATFORM

HIPAA-aware healthcare platform for real-time DICOM image sync across hospitals.

Category: fullstack

Tech: React.js, Firebase, MySQL, DICOM

ARCHITECTURE & RESULTS

STRUCTURED METADATA INDEXING

$O(\log n)$ lookups vs full-collection scans cut retrieval latency by 35%.

- Real-time DICOM sync across hospital nodes.
- Workflow digitization removed transfer bottlenecks.
- React.js frontend with Firebase + MySQL backend.

--- Page 2 of 2 ---